A Computer-Assisted Bound of Seven Singular Points for Rank-One klt del Pezzo Surfaces in Characteristic Three

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Abstract

We prove, by a corrected hybrid argument combining exact lattice computation and geometric elimination, that a rank-one klt del Pezzo surface over an algebraically closed field of characteristic 3 has at most seven singular points. The proof is by contradiction. Assuming $N(X) = 8$, we first carry out a computer-assisted basket reduction: after exact discriminant, odd-lattice, and anticanonical-complement screens, only five candidate baskets remain. Three baskets of type $7A_1+$ one exotic cyclic quotient singularity are then excluded by Weyl normalization of the exotic block together with a sharp bound on the orthogonal root subsystem. The mixed basket

$$\mathcal{B}_{\mathrm{mix}} = \left\{\tfrac{1}{2}(1,1),\ 5\times\tfrac{1}{3}(1,1),\ 2\times\tfrac{1}{3}(1,2)\right\}$$

is reduced to two polarization orbits. One orbit is eliminated by an exact root-combinatorial obstruction. The surviving orbit is globally normalized to a single canonical model, reduced to seven grouped symmetry types, and each type is ruled out by a tangent-contact plus Bézout argument. Finally, the Gorenstein basket $8A_1$ is shown to force degree 1; the anticanonical model is a finite separable double cover of $Q \cong \mathbb{P}(1,1,2)$, and after resolving the vertex to $\mathbb{F}_2$ the branch divisor has the form $S + C$ with $C \in |3(S + 2F)|$ disjoint from the negative section. A direct node bound shows that such a branch divisor supports at most six nodes, excluding $8A_1$. Therefore $N(X) \leq 7$. The accompanying repository records the exact finite computations used in the proof.[1]

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[1] Reproducibility repository: `https://github.com/HiroSakuraba/DelPezzo7`.

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1 Introduction

Let k be an algebraically closed field of characteristic 3, and let X be a rank-one klt del Pezzo surface over k. Write $N(X)$ for the number of singular points of X. The goal of this paper is to prove the following theorem.

Theorem 1.1 (Main theorem)**.** *Let X be a rank-one klt del Pezzo surface over an algebraically closed field of characteristic* 3*. Then*

$$N(X) \leq 7.$$

Seven-point constructions already appear in the broader rank-one log del Pezzo literature; for example, Lacini proves sharp seven-point statements in characteristics different from 2 and 3; see Example 7.4 and the discussion around Corollary 1.2 of [5]. The Keel–McKernan construction in characteristic 2 produces arbitrarily many singular points, so the interest of the present result lies specifically in the intermediate characteristic 3. The point of the present paper is the characteristic-3 upper bound and, in particular, the closure of the remaining $N = 8$ possibilities.

Remark 1.2 (Sharpness)**.** The question of whether $N = 7$ is actually achieved by a rank-one klt del Pezzo surface in characteristic 3 is not addressed in this paper. The bound $N \leq 7$ is the only claim we make.

Proof outline. The proof is by contradiction: we assume $N(X) = 8$ and derive an impossibility in every case. The argument has three independent components.

(1) *Lattice-theoretic reduction.* The orbifold Noether formula, combined with discriminant and odd-lattice bilinear screens, reduces all possible 8-point baskets to five survivors. This is the only stage at which a broad computer-assisted enumeration is used.

(2) *Root combinatorics.* The three exotic baskets ($7A_1$ plus one cyclic quotient) and the mixed basket are eliminated by showing that the lattice of exceptional curves on the minimal resolution is too constrained to support enough disjoint (-2)-curves or irreducible noncanonical exceptional curves. The key tools are Weyl/Cremona normalization, exact orthogonal root-system computations, and a Bézout/tangent-contact argument for irreducibility.

(3) *Classical algebraic geometry.* The pure Gorenstein basket $8A_1$ is eliminated by the degree-one weak del Pezzo double-cover model: the anticanonical model is a finite separable double cover of $Q \cong \mathbb{P}(1,1,2)$, and after resolving the vertex to $\mathbb{F}_2$ the pulled-back branch divisor can support at most six nodes.

The proof is hybrid. The first step is a finite exact reduction of all possible 8-point baskets to five survivors; this is the only stage at which a broad search is used. Every later step is either a short exact lattice computation or a geometric contradiction. Because the proof chain underwent several corrections during the exploratory phase, we write the paper only around the *final corrected chain*; superseded branches and failed subarguments are omitted from the logical core and are retained only in the repository history.

The five surviving $N = 8$ baskets are

$$8A_1, \qquad 7A_1 + \tfrac{1}{7}(1,1), \qquad 7A_1 + \tfrac{1}{7}(1,2), \qquad 7A_1 + \tfrac{1}{11}(1,2),$$

and the mixed basket

$$\mathcal{B}_{\mathrm{mix}} = \left\{ \tfrac{1}{2}(1,1),\ 5 \times \tfrac{1}{3}(1,1),\ 2 \times \tfrac{1}{3}(1,2) \right\}.$$

The paper is organized around closing these five possibilities:

- In section 4, the three exotic baskets are excluded by normalizing the noncanonical block and bounding the size of the orthogonal root subsystem.
- In section 5, the mixed basket is reduced to a single polarization orbit, globally normalized, and then eliminated by a tangent-contact contradiction.
- In section 6, the pure Gorenstein basket $8A_1$ is reduced to the degree-1 weak del Pezzo model and excluded by a branch-node bound.

The accompanying repository is part of the proof architecture. It contains the exact scripts used to enumerate candidate baskets, compute discriminant and root data, and verify the finite orbit counts that appear in the paper. The computational steps are exact and finite; no floating-point numerics or random sampling enter the argument.

At the same time, the repository is intentionally broader than the logical core of the paper. It also preserves several exploratory branches—including ambient/deformation lanes, quasi-elliptic probes, wild α_3/Artin–Schreier searches, and the earlier matroid/circuit screening tools—because those files are useful for provenance and for future work, whether human or machine-assisted. None of that exploratory material is used as a theorem dependency unless it is explicitly cited below.

2 Preliminaries

Notation 2.1. Throughout, k is an algebraically closed field of characteristic 3. Let

$$f\colon Y \to X$$

be the minimal resolution of a rank-one klt del Pezzo surface X. For a cyclic quotient singularity we use the notation $\frac{1}{r}(1,q)$. Whenever we use a blowup model, we write

$$Y \cong \mathrm{Bl}_{p_1,\dots,p_n}\mathbb{P}^2$$

with basis $H, E_1, \dots, E_n$ and canonical class

$$K_Y = -3H + E_1 + \cdots + E_n.$$

If $D = dH - \sum m_i E_i$, then $D^2 = d^2 - \sum m_i^2$ and $K_Y \cdot D = -3d + \sum m_i$.

Proposition 2.2 (Basic rational-surface facts)**.** *With notation as above:*

(i) *Y is a smooth rational surface.*

(ii) *If $R \subset \mathrm{NS}(Y)$ is the lattice generated by the exceptional curves of f, then*

$$\rho(Y) = \rho(X) + \mathrm{rk}(R).$$

(iii) *If all singularities of X are Du Val, then f is crepant and $K_Y = f^*K_X$.*

(iv) *For every smooth rational surface Y,*

$$\rho(Y) = 10 - K_Y^2.$$

Proof. The rationality of Y is standard for the minimal resolution of a rank-one del Pezzo surface. The Picard-number formula is immediate from the decomposition of $\mathrm{NS}(Y)$ into the pullback of $\mathrm{NS}(X)$ and the exceptional lattice. Crepancy for Du Val singularities is classical surface theory. Finally, for a smooth rational surface,

$$\chi(\mathcal{O}_Y) = 1, \qquad K_Y^2 + \rho(Y) = 10,$$

by Noether's formula and the vanishing of $q(Y)$ and $p_g(Y)$. □

Remark 2.3. For later use we note that if X is Gorenstein with only Du Val singularities, then the graded anticanonical ring is preserved by the minimal resolution. The precise statement appears in proposition 6.6.

Lemma 2.4 (Cohomological vanishing for anti-pluricanonical divisors on degree-one weak del Pezzo surfaces)**.** *Let Y be a smooth weak del Pezzo surface of degree one over an algebraically closed field of characteristic different from 2. Then for every $m \geq 1$,*

$$H^1(Y, \mathcal{O}_Y(-mK_Y)) = 0.$$

In particular, $h^0(-mK_Y) = \chi(-mK_Y) = \frac{m(m+1)}{2} + 1$.

Proof. The proof is by induction on m.

Base case $m = 1$. By Riemann–Roch on the smooth rational surface Y,

$$\chi(-K_Y) = \frac{(-K_Y)^2 - K_Y \cdot (-K_Y)}{2} + \chi(\mathcal{O}_Y) = \frac{1+1}{2} + 1 = 2.$$

By Serre duality, $H^2(Y, \mathcal{O}_Y(-K_Y)) \cong H^0(Y, \mathcal{O}_Y(2K_Y))^\vee$. Since $-K_Y$ is nef and $(2K_Y) \cdot (-K_Y) = -2 < 0$, the divisor $2K_Y$ is not effective, so $h^2(-K_Y) = 0$. Hence $h^0(-K_Y) - h^1(-K_Y) = 2$ and $h^0(-K_Y) \geq 2$.

For the reverse bound, consider the exact sequence

$$0 \to \mathcal{O}_Y \to \mathcal{O}_Y(-K_Y) \to \mathcal{O}_D(-K_Y|_D) \to 0$$

for any effective divisor $D \in |-K_Y|$. The curve D has arithmetic genus $p_a(D) = 1$ by adjunction: $(K_Y + D) \cdot D = 0$. The restriction $-K_Y|_D$ has degree $(-K_Y) \cdot D = 1$, so $h^0(D, \mathcal{O}_D(-K_Y|_D)) \leq 1$

by Riemann–Roch on the Gorenstein curve D (whether D is smooth, nodal, cuspidal, or reducible). Since $H^1(Y, \mathcal{O}_Y) = 0$ for the rational surface Y, the long exact sequence gives

$$0 = H^1(Y, \mathcal{O}_Y) \to H^1(Y, \mathcal{O}_Y(-K_Y)) \to H^1(D, \mathcal{O}_D(-K_Y|_D)).$$

It therefore suffices to show $H^1(D, \mathcal{O}_D(-K_Y|_D)) = 0$. By Serre duality on the Gorenstein curve D with $\omega_D \cong \mathcal{O}_D$ (from adjunction), one has $H^1(D, \mathcal{O}_D(-K_Y|_D)) \cong H^0(D, \mathcal{O}_D(K_Y|_D))^\vee$. But $K_Y|_D$ has degree $K_Y \cdot D = -1 < 0$, so $h^0(D, \mathcal{O}_D(K_Y|_D)) = 0$. Therefore $h^1(-K_Y) = 0$.

Inductive step. Fix any effective divisor $D \in |-K_Y|$. The curve D is a Gorenstein curve of arithmetic genus 1 with $\omega_D \cong \mathcal{O}_D$, but D need not be smooth (in characteristic 3 the generic member could be cuspidal if the pencil is quasi-elliptic). This does not affect the argument: we only use Serre duality on the Gorenstein curve D. Consider the exact sequence

$$0 \to \mathcal{O}_Y(-(m-1)K_Y) \to \mathcal{O}_Y(-mK_Y) \to \mathcal{O}_D(-mK_Y|_D) \to 0.$$

The restriction $-mK_Y|_D$ has degree $(-mK_Y) \cdot (-K_Y) = m$. By Serre duality on D,

$$H^1(D, \mathcal{O}_D(-mK_Y|_D)) \cong H^0(D, \mathcal{O}_D(mK_Y|_D))^\vee.$$

Since $mK_Y|_D$ has degree $-m < 0$, we get $h^0(D, \mathcal{O}_D(mK_Y|_D)) = 0$, hence $H^1(D, \mathcal{O}_D(-mK_Y|_D)) = 0$. The long exact sequence in cohomology gives

$$H^1(Y, \mathcal{O}_Y(-(m-1)K_Y)) \to H^1(Y, \mathcal{O}_Y(-mK_Y)) \to H^1(D, \mathcal{O}_D(-mK_Y|_D)) = 0.$$

By the inductive hypothesis, $H^1(Y, \mathcal{O}_Y(-(m-1)K_Y)) = 0$. Therefore $H^1(Y, \mathcal{O}_Y(-mK_Y)) = 0$.

The formula $h^0 = \chi$ then follows from $H^2(-mK_Y) \cong H^0((m+1)K_Y)^\vee = 0$. □

3 Reduction to five candidate baskets

This section is the only place where the proof uses a broad finite computer-assisted search. The underlying computation is exact. It proceeds in four stages:

(1) enumerate all local cyclic quotient types that survive the elementary $N = 8$ local constraints,

(2) enumerate all 8-point multisets built from those local types,

(3) impose orbifold and discriminant constraints,

(4) replace the provisional quadratic-form screen by the corrected odd-lattice screen on the discriminant bilinear form and anticanonical complement.

The output is the five-basket list of proposition 3.1.

Proposition 3.1 (Final $N = 8$ basket reduction)**.** *Assume X is a rank-one klt del Pezzo surface over k with $N(X) = 8$. Then the singularity basket of X is one of the following five baskets:*

$$8A_1, \qquad 7A_1 + \tfrac{1}{7}(1,1), \qquad 7A_1 + \tfrac{1}{7}(1,2), \qquad 7A_1 + \tfrac{1}{11}(1,2),$$

and

$$\mathcal{B}_{\mathrm{mix}} = \left\{ \tfrac{1}{2}(1,1),\ 5 \times \tfrac{1}{3}(1,1),\ 2 \times \tfrac{1}{3}(1,2) \right\}.$$

Proof. We summarize the exact finite reduction; the full local-type table, scripts, and machine-readable outputs appear in appendices A and C. The reduction is exact and finite at every stage.

The local screen leaves 17 cyclic quotient types, namely

$$\tfrac{1}{2}(1,1),\ \tfrac{1}{3}(1,1),\ \tfrac{1}{4}(1,1),\ \tfrac{1}{5}(1,1),\ \tfrac{1}{6}(1,1),\ \tfrac{1}{7}(1,1),\ \tfrac{1}{8}(1,1),\ \tfrac{1}{9}(1,1),\ \tfrac{1}{10}(1,1),\ \tfrac{1}{11}(1,1),$$
$$\tfrac{1}{3}(1,2),\ \tfrac{1}{5}(1,2),\ \tfrac{1}{8}(1,3),\ \tfrac{1}{7}(1,2),\ \tfrac{1}{11}(1,3),\ \tfrac{1}{9}(1,2),\ \tfrac{1}{11}(1,2).$$

Exact enumeration of all 8-point multisets built from these types gives 735,471 baskets. The successive exact reductions are summarized in table 1.

stage	number of baskets
all 8-point multisets from the 17 local types	735,471
positive candidate K_X^2	732,878
coarse square-discriminant budget	4,894
corrected exact small-discriminant window (resolved)	45
final survivors after odd-lattice bilinear / anticanonical screen	5

Table 1: Exact basket-reduction counts used in proposition 3.1.

In the small-discriminant window, the corrected exact search decides 45 baskets immediately: 24 pass the preliminary exact discriminant test and 21 fail. Two additional historically unresolved baskets were then resolved negatively later in the corrected pipeline. After replacing the provisional quadratic-form screen by the corrected odd-lattice bilinear/anticanonical screen, exactly the five baskets listed above remain.

The proof is exact and finite. The scripts, machine-readable outputs, and intermediate counts are recorded in the repository and summarized in appendices A and C. □

Lemma 3.2 (How the numerical basket screens are formed)**.** *Let $\mathcal{B} = \{P_1, \dots, P_8\}$ be an 8-point basket drawn from the local types of table 2. For each point P_i, let r_i be the rank of the exceptional chain on the minimal resolution and let $\delta_i := K_X^2 - K_Y^2$ be the local discrepancy correction recorded in table 2. Then any rank-one klt del Pezzo surface with basket $\mathcal{B}$ must satisfy*

$$K_X^2 = 9 - \sum_{i=1}^{8} r_i + \sum_{i=1}^{8} \delta_i > 0.$$

Moreover, if R denotes the total exceptional lattice and m is any positive integer for which mK_X is Cartier, then the discriminant compatibility for the rank-one complement in $\mathrm{NS}(Y)$ *imposes the square condition*

$$|A_R|\, m^2 K_X^2 \in (\mathbf{Z}_{>0})^2,$$

and the refined odd-lattice screen requires that the discriminant bilinear form of R admit a cyclic rank-one complement carrying the anticanonical class of the required norm.

Proof. Since $\rho(X) = 1$ and the total exceptional rank is $r := \sum r_i$, one has

$$\rho(Y) = 1 + r.$$

Because Y is smooth and rational, proposition 2.2(iv) gives

$$K_Y^2 = 10 - \rho(Y) = 9 - r.$$

By definition, each local correction δ_i satisfies $K_X^2 - K_Y^2 = \delta_i$ across the corresponding singularity, so summing over the basket gives

$$K_X^2 = K_Y^2 + \sum_{i=1}^{8} \delta_i = 9 - \sum r_i + \sum \delta_i.$$

Positivity of $-K_X$ forces $K_X^2 > 0$, giving the first screen.

For the second statement, write $\bar{R} \subset \mathrm{NS}(Y)$ for the primitive closure of the exceptional lattice. Since $\mathrm{NS}(Y)$ is an odd unimodular lattice of signature $(1, \rho(Y) - 1)$, a rank-one complement generated by a class h of square $h^2 = m^2 K_X^2$ can exist only if the discriminant group of $\bar{R} \oplus \mathbf{Z}h$ admits an isotropic glue to a unimodular overlattice. In particular the product of discriminants must be a square, which yields the displayed necessary condition. The corrected final screen is the exact version of this statement: one works prime-by-prime with the discriminant bilinear form of $\bar{R}$ and asks whether there is a cyclic rank-one complement carrying the anticanonical class of the required norm. This is the odd-lattice bilinear/anticanonical test implemented in the repository. □

Remark 3.3 (Computer-assisted status)**.** Proposition 3.1 is the only proposition in the paper that depends on a broad exact enumeration rather than a short intrinsic geometric argument. All later eliminations are either hand proofs or exact finite checks attached to a single already-normalized configuration.

4 The three exotic baskets

We first eliminate the three baskets

$$7A_1 + \tfrac{1}{7}(1,1), \qquad 7A_1 + \tfrac{1}{7}(1,2), \qquad 7A_1 + \tfrac{1}{11}(1,2).$$

4.1 Normalization of the noncanonical block

Proposition 4.1 (Standard-form normalization)**.** *Let $Y \to X$ be the minimal resolution of one of the three exotic baskets above. Then, up to Weyl/Cremona action, the noncanonical exceptional block admits a unique standard representative:*

$$\begin{aligned}
\tfrac{1}{7}(1,1): &\quad C \sim H - E_1 - \cdots - E_8, \\
\tfrac{1}{7}(1,2): &\quad C_2 = E_5 - E_6, \qquad C_1 \sim H - E_1 - \cdots - E_5, \\
\tfrac{1}{11}(1,2): &\quad C_2 = E_7 - E_8, \qquad C_1 \sim H - E_1 - \cdots - E_7.
\end{aligned}$$

Here the chains are ordered so that the (-2)-component is denoted by C_2 whenever the exotic point resolves to a two-curve chain.

Proof. We solve the standard-form equations in each case.

Case $7A_1 + \frac{1}{7}(1,1)$. The noncanonical exceptional curve C satisfies

$$C^2 = -7, \qquad K_Y \cdot C = 5.$$

The total exceptional rank is $7 \cdot 1 + 1 = 8$ (seven A_1-curves of rank 1 each, plus the single noncanonical curve), so $\rho(Y) = 1 + 8 = 9$ and $K_Y^2 = 10 - 9 = 1$. Write

$$C = dH - \sum_{i=1}^{8} m_i E_i$$

in standard form with $d \geq m_1 \geq \cdots \geq m_8 \geq 0$. Then

$$\sum m_i = 3d + 5, \qquad \sum m_i^2 = d^2 + 7.$$

By Cauchy,

$$(3d+5)^2 \leq 8(d^2+7),$$

which forces $d \leq 1$. Since $d = 0$ is impossible for an effective standard-form class of self-intersection -7, we get $d = 1$. Then

$$\sum m_i = 8, \qquad \sum m_i^2 = 8,$$

so necessarily $m_1 = \cdots = m_8 = 1$.

Case $7A_1 + \frac{1}{7}(1,2)$. The exceptional chain is $[4,2]$, which has rank 2. The total exceptional rank is $7 \cdot 1 + 2 = 9$, giving $\rho(Y) = 10$ and $K_Y^2 = 0$, so $Y = \mathrm{Bl}_9\mathbb{P}^2$. The (-2)-curve C_2 is a real root in the affine E_8 lattice $K_Y^{\perp}/\mathbb{Z}K_Y$, so the Weyl group is transitive on its root orbit. We may therefore normalize

$$C_2 = E_5 - E_6.$$

Write

$$C_1 = dH - \sum_{i=1}^{9} m_i E_i$$

in standard form. Then

$$\sum m_i = 3d + 2, \qquad \sum m_i^2 = d^2 + 4, \qquad m_5 - m_6 = 1.$$

The exact standard-form solve for the first two equations has only two solutions:

$$(1; 1,1,1,1,1,1,0,0,0,0), \qquad (2; 1,1,1,1,1,1,1,1,1,0).$$

The second violates $m_5 - m_6 = 1$, so the first is the unique admissible solution. Hence

$$C_1 \sim H - E_1 - \cdots - E_5.$$

Case $7A_1 + \frac{1}{11}(1,2)$. The exceptional chain is $[6,2]$, which has rank 2. The total exceptional rank is $7 \cdot 1 + 2 = 9$, giving $\rho(Y) = 10$ and $K_Y^2 = 0$, so again $Y = \mathrm{Bl}_9\mathbb{P}^2$. We normalize the root component to

$$C_2 = E_7 - E_8.$$

For the other component,

$$\sum m_i = 3d + 4, \qquad \sum m_i^2 = d^2 + 6, \qquad m_7 - m_8 = 1.$$

The exact reduced solve gives the unique solution

$$(1; 1,1,1,1,1,1,1,0,0),$$

hence

$$C_1 \sim H - E_1 - \cdots - E_7.$$

This proves the proposition. $\square$

4.2 The orthogonal root subsystems

Lemma 4.2 (Orthogonal real-root sets for the exotic baskets)**.** *In the three normal forms of proposition 4.1, the orthogonal real-root sets have cardinalities*

$$56, \qquad 48, \qquad 32.$$

More precisely:

(i) *for* $\frac{1}{7}(1,1)$ *the orthogonal root subsystem is of type* A_7*;*

(ii) *for* $\frac{1}{7}(1,2)$ *the orthogonal real-root set has* 48 *elements and contains at most* 4 *pairwise orthogonal roots;*

(iii) *for* $\frac{1}{11}(1,2)$ *the orthogonal real-root set has* 32 *elements and contains at most* 4 *pairwise orthogonal roots.*

In particular, the maximal number of pairwise orthogonal roots is at most

$$4, \qquad 4, \qquad 4.$$

Proof. We record both the structural picture and the exact finite verification.

For $\frac{1}{7}(1,1)$, the orthogonal roots to

$$C = H - E_1 - \cdots - E_8$$

inside $K_Y^{\perp} \cong E_8(-1)$ contain the chain

$$E_1 - E_2,\ E_2 - E_3,\ \ldots,\ E_7 - E_8,$$

which gives a root subsystem of type A_7. The exact root enumeration in the repository returns exactly 56 orthogonal real roots, which is the full root count of A_7; hence the orthogonal root subsystem is exactly A_7. In particular the maximal number of pairwise orthogonal roots is 4.

For $\frac{1}{7}(1,2)$, after normalizing

$$C_2 = E_5 - E_6, \qquad C_1 = H - E_1 - \cdots - E_5,$$

the orthogonal real-root set in $K_Y^{\perp}$ can be described explicitly as the union of

- the 12 permutation roots on $\{E_1, E_2, E_3, E_4\}$,
- the 6 permutation roots on $\{E_7, E_8, E_9\}$,
- and 15 roots of the form $H - E_a - E_b - E_c$ together with their negatives,

for a total of $12+6+30 = 48$ roots. An exact clique computation on the orthogonality graph shows that the largest pairwise orthogonal subset has size 4.

For $\frac{1}{11}(1,2)$, after normalizing

$$C_2 = E_7 - E_8, \qquad C_1 = H - E_1 - \cdots - E_7,$$

the orthogonal real-root set consists of the 30 permutation roots on $\{E_1, \ldots, E_6\}$ together with the two roots

$$\pm(H - E_7 - E_8 - E_9),$$

for a total of 32 roots. Again an exact clique computation on the orthogonality graph shows that the maximum number of pairwise orthogonal roots is 4. □

Theorem 4.3 (Exotic baskets are impossible)**.** *None of the three baskets*

$$7A_1 + \tfrac{1}{7}(1,1), \qquad 7A_1 + \tfrac{1}{7}(1,2), \qquad 7A_1 + \tfrac{1}{11}(1,2)$$

occur on a rank-one klt del Pezzo surface in characteristic 3*.*

Proof. Each basket would require seven further disjoint A_1-curves orthogonal to the noncanonical block. By lemma 4.2, the orthogonal real-root subsystem is too small in every case: it supports at most 4, 4, and 4 pairwise orthogonal roots respectively. This contradiction rules out all three baskets. □

5 The mixed basket

We now eliminate the mixed basket

$$\mathcal{B}_{\mathrm{mix}} = \left\{ \tfrac{1}{2}(1,1),\ 5 \times \tfrac{1}{3}(1,1),\ 2 \times \tfrac{1}{3}(1,2) \right\}.$$

The proof is longer, but every step after the initial basket reduction is finite and exact.

5.1 Polarization reduction

Proposition 5.1 (The mixed-basket polarization)**.** *Assume X has basket $\mathcal{B}_{\mathrm{mix}}$, and let $f\colon Y \to X$ be the minimal resolution. If $C_1, \dots, C_5$ are the five noncanonical (-3)-curves coming from the five $\frac{1}{3}(1,1)$ points, then*

$$L := 3f^*(-K_X) = -3K_Y - \sum_{i=1}^{5} C_i$$

is an integral nef divisor satisfying

$$L^2 = 6, \qquad K_Y \cdot L = -2.$$

After Cremona reduction, L lies in one of exactly two standard-form orbits:

$$L_1 = 4H - E_1 - \cdots - E_{10}, \qquad L_2 = 6H - 2E_1 - \cdots - 2E_7 - E_8 - E_9.$$

Proof. For a $\frac{1}{3}(1,1)$ point, the discrepancy on the exceptional curve is $-\frac{1}{3}$, while the singularities A_1 and $\frac{1}{3}(1,2)$ are canonical. Thus

$$K_Y = f^*K_X - \frac{1}{3}\sum_{i=1}^{5} C_i.$$

Rearranging gives the stated integral divisor L. Since $f^*(-K_X)$ is nef and big, so is L. Direct intersection computation yields

$$L^2 = 9K_Y^2 + 6\sum K_Y \cdot C_i + \sum_{i,j} C_i \cdot C_j = 6, \qquad K_Y \cdot L = -3K_Y^2 - \sum K_Y \cdot C_i = -2.$$

Write

$$L = dH - \sum_{i=1}^{10} m_i E_i$$

in standard form. Then

$$\sum m_i = 3d - 2, \qquad \sum m_i^2 = d^2 - 6.$$

A finite reduced standard-form solve gives exactly two solutions up to Cremona equivalence, namely L_1 and L_2. □

Proposition 5.2 (The polarization orbit L_1 is impossible)**.** *The mixed basket cannot occur in the polarization orbit L_1.*

Proof. Define the root sets

$$\mathcal{R}_0(L) = \{r \mid r^2 = -2,\ K_Y \cdot r = 0,\ L \cdot r = 0\}, \qquad \mathcal{R}_{-2}(L) = \{r \mid r^2 = -2,\ K_Y \cdot r = 0,\ L \cdot r = -2\}.$$

For $L = L_1$, the exact search gives

$$|\mathcal{R}_0(L_1)| = 90, \qquad |\mathcal{R}_{-2}(L_1)| = 210.$$

A mixed-basket realization would require a 5-clique in $\mathcal{R}_{-2}(L_1)$ with pairwise intersections 1, together with an orthogonal canonical subsystem of type $A_2 + A_2 + A_1$ in $\mathcal{R}_0(L_1)$. The graph on $\mathcal{R}_{-2}(L_1)$ contains exactly 30,240 such transformed noncanonical five-cliques, but the exact compatibility search finds that none is orthogonal to an $A_2+A_2+A_1$ subsystem. Hence L_1 is impossible. □

Thus any mixed-basket realization must occur in the single surviving orbit L_2.

5.2 Global normalization in the L_2 orbit

Theorem 5.3 (Global normalization theorem for the surviving L_2 orbit)**.** *Inside the L_2 root-theoretic model, canonical subsystems of type $A_2 + A_2 + A_1$ split into exactly two Weyl orbits, of sizes*

$$3360 \qquad \textit{and} \qquad 10080.$$

Among these two orbits, only the orbit of size 3360 *is compatible with a transformed noncanonical* 5*-clique. Consequently, every mixed-basket lattice configuration in the L_2 orbit is Weyl-equivalent to a canonical subsystem in the compatible* 3360*-orbit. For the grouped geometric analysis below we fix the convenient representative*

$$S_{\mathrm{std}} = \langle E_3 - E_1,\ E_1 - E_2,\ E_6 - E_4,\ E_5 - E_6,\ E_7 - E_8 \rangle.$$

Proof. Set

$$\mathcal{R}_0 = \mathcal{R}_0(L_2), \qquad \mathcal{R}_{-2} = \mathcal{R}_{-2}(L_2).$$

The exact root search gives

$$|\mathcal{R}_0| = 128, \qquad |\mathcal{R}_{-2}| = 177.$$

Now enumerate all canonical subsystems of type $A_2 + A_2 + A_1$ inside $\mathcal{R}_0$. There are exactly

$$13{,}440$$

such rootsets. Under the Weyl group generated by reflections in the roots of $\mathcal{R}_0$, these split into two orbits of sizes 3360 and 10080.

Next enumerate all transformed noncanonical five-cliques inside $\mathcal{R}_{-2}$, *i.e.* all 5-subsets with pairwise intersection 1. There are exactly

$$5600$$

such cliques. For each one, enumerate all orthogonal canonical subsystems of type $A_2 + A_2 + A_1$. The result is uniform: every transformed noncanonical five-clique is compatible only with the orbit of size 3360, namely the orbit containing S_{std}. None is compatible with the second canonical orbit.

Therefore every full mixed-basket lattice configuration can be moved by a $W(L_2)$-isometry so that its canonical subsystem is exactly S_{std}. □

5.3 Grouped reduction

Proposition 5.4 (Grouped reduction in the normalized L_2 model)**.** *After the normalization of theorem 5.3, every noncanonical class in the mixed-basket configuration has grouped form*

$$D = dH - a(E_1 + E_2 + E_3) - b(E_4 + E_5 + E_6) - c(E_7 + E_8) - mE_9 - nE_{10},$$

with multiplicity equalities

$$m_1 = m_2 = m_3, \qquad m_4 = m_5 = m_6, \qquad m_7 = m_8.$$

Up to the natural symmetries exchanging the two length-3 clusters and the two terminal points, the resulting pairwise-orthogonal transformed noncanonical five-cliques fall into exactly seven orbit types.

Proof. Orthogonality to the five roots spanning S_{std} forces the displayed equalities:

$$(E_3 - E_1) \cdot D = 0 \Rightarrow m_1 = m_3, \qquad (E_1 - E_2) \cdot D = 0 \Rightarrow m_1 = m_2,$$

$$(E_6 - E_4) \cdot D = 0 \Rightarrow m_4 = m_6, \qquad (E_5 - E_6) \cdot D = 0 \Rightarrow m_5 = m_6,$$

$$(E_7 - E_8) \cdot D = 0 \Rightarrow m_7 = m_8.$$

Thus every candidate noncanonical class is grouped. The exact clique enumeration in the normalized grouped model produces 16 grouped five-cliques; quotienting by the obvious symmetries leaves precisely 7 orbit types. □

5.4 Tangent contact and the seven orbit types

Lemma 5.5 (Tangent-contact lemma)**.** *Let*

$$D = dH - a(E_1 + E_2 + E_3) - b(E_4 + E_5 + E_6) - c(E_7 + E_8) - mE_9 - nE_{10}$$

be an effective grouped class in the normalized model. Let T_A, T_B, T_C denote the tangent lines to the three proximate clusters of lengths $3, 3, 2$ respectively. Then

$$I(T_A, D) \geq 3a, \qquad I(T_B, D) \geq 3b, \qquad I(T_C, D) \geq 2c.$$

Consequently, if any one of the inequalities

$$3a > d, \qquad 3b > d, \qquad 2c > d$$

holds, then the corresponding tangent line is forced as a component by Bézout, and the strict transform of D on the blowup is reducible.

Proof. The argument is local and characteristic-free; it uses only the theory of infinitely near points in the plane (see *e.g.* [2, §3.3]). The length-3 multiplicity sequence (a, a, a) says that the plane curve representing D meets the tangent line T_A with contact order at least $3a$ after successively blowing up the three infinitely near points in the A-cluster: at each stage the strict transform passes through the exceptional point with multiplicity at least a, and the total contact order is the sum. The same argument gives contact order at least $3b$ with T_B and $2c$ with T_C. If one of these contact orders exceeds the plane degree d, then Bézout forces the corresponding tangent line to be an actual component of the plane curve. The strict transform is therefore reducible. □

Proposition 5.6 (The seven grouped L_2 orbit types are all impossible)**.** *For each of the seven grouped orbit types from proposition 5.4, one of the five classes in any representative grouped five-clique violates one of the inequalities in lemma 5.5. Consequently every grouped five-clique in each orbit type contains a reducible member, so none of the seven orbit types can realize five irreducible noncanonical exceptional* (-3)*-curves.*

Proof. For each orbit type we inspect a representative grouped five-clique and exhibit one of its five classes that is necessarily reducible. Write a grouped class as $(d; a, b, c; m, n)$. The exact orbit representatives are listed in the repository; the following table records, for each orbit type, a member of the representative five-clique that is forced reducible by lemma 5.5. Because the orbit type is an orbit of whole grouped five-cliques under the residual symmetry group, the existence of such a distinguished reducible member is invariant across the orbit type.

Orbit	representative $(d; a, b, c; m, n)$	forced inequality
1	$(5; 2, 1, 2; 1, 2)$	$3a = 6 > 5$
2	$(2; 1, 1, 0; 0, 1)$	$3a = 3 > 2$, $3b = 3 > 2$
3	$(2; 1, 1, 0; 0, 1)$	$3a = 3 > 2$, $3b = 3 > 2$
4	$(7; 3, 2, 2; 2, 1)$	$3a = 9 > 7$
5	$(5; 1, 2, 2; 2, 1)$	$3b = 6 > 5$
6	$(4; 1, 2, 1; 1, 1)$	$3b = 6 > 4$
7	$(4; 1, 2, 1; 1, 1)$	$3b = 6 > 4$

In every case the relevant tangent line is forced as a component by lemma 5.5, so the displayed member of the representative five-clique is reducible. Since grouped-orbit equivalence acts on whole five-cliques, every five-clique in the same orbit type contains a corresponding reducible member. Therefore no orbit type can realize the required five irreducible noncanonical exceptional curves. Note that orbits 2 and 3 share the same displayed forced-reducible class, as do orbits 6 and 7; the symmetry orbits are different five-cliques, but one may choose the same witness class in each pair. □

Theorem 5.7 (The mixed basket is impossible)**.** *The mixed basket*

$$\mathcal{B}_{\mathrm{mix}} = \left\{\tfrac{1}{2}(1,1),\ 5 \times \tfrac{1}{3}(1,1),\ 2 \times \tfrac{1}{3}(1,2)\right\}$$

does not occur on a rank-one klt del Pezzo surface in characteristic 3*.*

Proof. By proposition 5.1, any realization of the mixed basket lies in one of the polarization orbits L_1 or L_2. The orbit L_1 is impossible by proposition 5.2. Thus only L_2 remains.

By theorem 5.3, every full L_2 configuration is Weyl-equivalent to the standard canonical subsystem S_{std}. By proposition 5.4, every transformed noncanonical five-clique is then one of the seven grouped orbit types. By proposition 5.6, every such grouped five-clique contains a reducible member, contradicting irreducibility of the five noncanonical exceptional curves. Therefore the mixed basket cannot occur. □

6 The pure $8A_1$ case

We now eliminate the remaining Gorenstein basket $8A_1$.

6.1 Degree one and the weak del Pezzo model

Proposition 6.1 (The pure $8A_1$ case has degree one)**.** *Assume X has basket $8A_1$. Then*

$$K_X^2 = 1.$$

Proof. Let $f\colon Y \to X$ be the minimal resolution. Since the singularities are all Du Val of type A_1, the resolution is crepant, so

$$K_Y = f^*K_X, \qquad K_Y^2 = K_X^2 =: d.$$

Also

$$\rho(Y) = \rho(X) + 8 = 9.$$

By proposition 2.2(iv),

$$10 - K_Y^2 = \rho(Y) = 9,$$

so $d = 1$. □

Definition 6.2 (Almost general position)**.** A set of eight points in $\mathbb{P}^2$ is said to be in *almost general position* if no three are collinear, no six lie on a conic, and no cubic passes through all eight with one of them as a singular point. Equivalently, the blowup has nef anticanonical class.

Remark 6.3 (Almost general position versus the presence of (-2)-curves)**.** The phrase "almost general position" in proposition 6.4 is the standard weak-del-Pezzo condition: it ensures that $-K_Y$ is nef. It does *not* mean that the eight points are generic in the everyday sense. In particular, the existence of (-2)-curves on a weak del Pezzo surface is compatible with almost general position; these curves record degeneracies in the blowup configuration that are still allowed by the nefness of $-K_Y$.

Proposition 6.4 (Degree-one weak del Pezzo structure)**.** *Let Y be a weak del Pezzo surface of degree one over an algebraically closed field of characteristic 3. Then:*

(i) $Y \cong \mathrm{Bl}_{p_1,\dots,p_8}\mathbb{P}^2$ *for eight points in almost general position;*

(ii) $|-K_Y|$ *is a pencil with exactly one base point q;*

(iii) *after blowing up q, one obtains a smooth rational surface*

$$\sigma\colon Z := \mathrm{Bl}_q Y \to Y$$

whose anticanonical pencil induces a relatively minimal genus-one fibration $Z \to \mathbb{P}^1$;

(iv) *the linear system $|-2K_Y|$ is basepoint-free and induces a separable double cover of the quadric cone.*

Proof. Statement (i) is the standard degree-one weak-del-Pezzo classification: a weak del Pezzo surface of degree one is the blowup of $\mathbb{P}^2$ at eight points in almost general position; see Demazure [3]. Statement (ii) is the classical description of the anti-canonical system in degree one: $|-K_Y|$ is a genus-one pencil with a unique base point. Indeed, by Riemann–Roch $\chi(-K_Y) = 2$, and the vanishing $H^1(Y, \mathcal{O}_Y(-K_Y)) = 0$ (from lemma 2.4) gives $h^0(-K_Y) = 2$. Blowing up that base point gives the smooth rational surface Z together with the induced relatively minimal genus-one fibration $Z \to \mathbb{P}^1$ of (iii), which in characteristic 3 may be elliptic or quasi-elliptic. Statement (iv) is the bi-anticanonical model theorem in characteristic different from 2: the linear system $|-2K_Y|$ is basepoint-free and induces a separable double cover of the quadric cone. For normal Gorenstein

degree-one del Pezzo surfaces this is recorded by Hidaka–Watanabe [4]; for degree-one del Pezzo surfaces with canonical singularities the same double-cover description appears in Proposition 2.2 and Remark 2.3 of Ascher–Bejleri [1]. The basepoint-freeness of $|-2K_Y|$ on a weak del Pezzo of degree one also uses $H^1(Y, \mathcal{O}_Y(-2K_Y)) = 0$, which follows from lemma 2.4. The separability in characteristic 3 is immediate because the covering degree is 2. □

Lemma 6.5 (The anti-canonical base point avoids the (-2)-curves)**.** *Let Y be a degree-one weak del Pezzo surface, and let q be the unique base point of $|-K_Y|$. Then q lies on no (-2)-curve of Y.*

Proof. Let $R \subset Y$ be a (-2)-curve. Since $-K_Y$ is nef and R is orthogonal to the anti-canonical class, one has

$$(-K_Y) \cdot R = 0.$$

Assume for contradiction that $q \in R$. Every divisor $D \in |-K_Y|$ passes through q, so the effective divisors D and R meet. Because $D \cdot R = 0$, positivity of local intersection numbers forces R to be a component of every D. Hence R is a fixed component of $|-K_Y|$.

Write

$$|-K_Y| = R + |M|.$$

Then $M \sim -K_Y - R$, and since $h^0(-K_Y) = 2$ (by Riemann–Roch and lemma 2.4, as $-K_Y$ is nef), the moving part $|M|$ is nonempty. But

$$M^2 = (-K_Y - R)^2 = K_Y^2 + R^2 - 2(-K_Y \cdot R) = 1 - 2 - 0 = -1,$$

which is impossible for the moving part of a nontrivial pencil. Therefore $q \notin R$. □

Proposition 6.6 (Equality of anticanonical rings under the crepant resolution)**.** *Let X be a Gorenstein del Pezzo surface with only Du Val singularities, and let $f\colon Y \to X$ be its minimal resolution. Then for every $m \geq 0$,*

$$H^0(X, \mathcal{O}_X(-mK_X)) \cong H^0(Y, \mathcal{O}_Y(-mK_Y)),$$

so

$$R(X, -K_X) \cong R(Y, -K_Y).$$

Proof. Du Val singularities are rational, so $R^i f_* \mathcal{O}_Y = 0$ for $i > 0$ and $f_* \mathcal{O}_Y = \mathcal{O}_X$. Since f is crepant, $K_Y = f^* K_X$, hence

$$\mathcal{O}_Y(-mK_Y) \cong f^* \mathcal{O}_X(-mK_X)$$

for every $m \geq 0$. By the projection formula,

$$f_* \mathcal{O}_Y(-mK_Y) \cong \mathcal{O}_X(-mK_X) \otimes f_* \mathcal{O}_Y \cong \mathcal{O}_X(-mK_X).$$

Taking global sections proves the claim. □

6.2 The separable branch model

The key structural point is that the anticanonical model of X (equivalently of Y, by proposition 6.6) is a finite separable double cover of the quadric cone. To apply the node correspondence, we work with this finite cover directly, not with the smooth blowup Z.

Proposition 6.7 (The anticanonical model as a finite double cover)**.** *Assume X has basket $8A_1$, and let $f\colon Y \to X$ be the minimal resolution. Then the morphism induced by $|-2K_Y|$ factors as*

$$Y \xrightarrow{g} X \xrightarrow{\pi} Q \cong \mathbb{P}(1,1,2),$$

where $g = f$ is the minimal resolution and π is a finite separable double cover. After resolving the vertex of Q to obtain $\rho\colon \mathbb{F}_2 \to Q$, the normalized fiber product gives the commutative diagram

$$\begin{tikzcd} Z \arrow[r, "\sigma"] \arrow[d, "\widetilde{\pi}"'] & Y \arrow[r, "f"] & X \arrow[d, "\pi"] \\ \mathbb{F}_2 \arrow[rr, "\rho"] & & Q \end{tikzcd}$$

where $Z = \mathrm{Bl}_q Y$ is smooth, $\widetilde{\pi}\colon Z \to \mathbb{F}_2$ is a generically finite morphism of degree 2 (but not finite: it contracts the strict transforms of the eight (-2)-curves to points), and $\pi\colon X \to Q$ is the finite separable double cover branched along a weighted sextic.

On $\mathbb{F}_2$, the branch divisor of the finite cover π pulls back to

$$B = S + C,$$

where S is the negative section and $C \in |3M|$, $M := S + 2F$, $C \cdot S = 0$.

Proof. By proposition 6.4(iv), $|-2K_Y|$ is basepoint-free and induces a separable morphism of degree 2 to $Q \cong \mathbb{P}(1,1,2)$. By proposition 6.6, $R(X, -K_X) \cong R(Y, -K_Y)$, so the anticanonical model

$$X \cong \mathrm{Proj} R(Y, -K_Y)$$

is the Stein factorization of the $|-2K_Y|$-map: the morphism $Y \to Q$ factors as $Y \xrightarrow{f} X \xrightarrow{\pi} Q$, where f contracts the eight (-2)-curves and π is finite of degree 2.

The branch divisor computation on $\mathbb{F}_2$ follows from the standard theory of degree-one del Pezzo surfaces [1, Prop. 2.2], [4]: after resolving the vertex of Q via $\rho\colon \mathbb{F}_2 \to Q$, the branch locus of π pulls back to $B = S + C$ with $C \in |3M|$, $C \cdot S = 0$. The exceptional curve E_q of $\sigma\colon Z \to Y$ is a section of the genus-one fibration on Z and maps to S on $\mathbb{F}_2$; it is pointwise fixed by the Bertini involution, hence S is a branch component. The class computation

$$S \cdot C = S \cdot 3M = 3(S^2 + 2S \cdot F) = 3(-2+2) = 0, \qquad B \sim S + 3M = 4S + 6F = 2(2S + 3F)$$

verifies disjointness and the even branch class. □

Lemma 6.8 (Local node correspondence for separable double covers)**.** *Let U be a smooth surface over a field of characteristic different from 2, and let*

$$\pi\colon V \to U$$

be a finite separable double cover branched along a reduced divisor $B \subset U$. Let $p \in B$ be a point where U is smooth and B has an ordinary node. Then the unique point of V lying over p is a Du Val singularity of type A_1. Conversely, if B is reduced and the point of V over p is of type A_1, then B has an ordinary node at p.

Proof. Choose local coordinates (x, y) on U centered at p. Because π is separable and 2 is invertible, the cover is analytically given by an equation

$$z^2 = f(x, y)$$

with branch divisor $f = 0$. If B has a node at p, then after a formal change of coordinates,

$$f(x, y) = xy + (\text{higher order terms})$$

with nondegenerate quadratic part. Completing the square and absorbing higher-order units, the singularity is analytically equivalent to

$$z^2 = xy,$$

which is the standard A_1 singularity. The converse follows from the same normal form: an A_1 point of the cover is analytically equivalent to $z^2 = uv$, and projecting to the base recovers a reduced branch divisor with ordinary quadratic part $uv = 0$, hence an ordinary node. □

Remark 6.9. In our situation the finite double cover is $\pi\colon X \to Q$, not the morphism from the smooth surface Z. The eight A_1-points of X lie over the smooth locus of Q (and away from the vertex, since the corresponding (-2)-curves on Y avoid the anticanonical base point by lemma 6.5). On $\mathbb{F}_2$, their images lie away from S and in the smooth locus of the ambient surface, so locally the branch equation is determined by C alone. By lemma 6.8, the eight A_1-points of X correspond to eight ordinary nodes of C.

6.3 The branch-node bound

Lemma 6.10 (Component partition for divisors in $|3M|$ disjoint from S)**.** *Let $C \in |3M|$ be a reduced effective divisor on $\mathbb{F}_2$ with $C \cdot S = 0$. Then every irreducible component of C has class aM for some positive integer a, and the multiplicities form a partition of* 3.

Proof. Let D be an irreducible component of C. Write $D \sim \alpha S + \beta F$ with $\alpha \geq 0$. Since D is effective and irreducible with $D \neq S$, one has $D \cdot S = \beta - 2\alpha \geq 0$. But D is a component of C and $C \cdot S = 0$, so $D \cdot S \geq 0$. Since every component of C satisfies $D \cdot S \geq 0$ and the total intersection is $C \cdot S = 0$, each component satisfies $D \cdot S = 0$. Hence $\beta = 2\alpha$ and $D \sim \alpha(S + 2F) = \alpha M$. The components partition the class $3M$, giving a partition of 3. □

Lemma 6.11 (Node bound for reduced divisors in $|3M|$ on $\mathbb{F}_2$)**.** *Let $C \in |3M|$ be a reduced divisor on $\mathbb{F}_2$, where $M = S + 2F$. Then C has at most six ordinary nodes.*

Proof. By lemma 6.10, the irreducible components of C partition the class $3M$ into multiples of M, so only the partitions 3, $2 + 1$, or $1 + 1 + 1$ of the coefficient arise. Since

$$M^2 = 2, \qquad p_a(M) = 0, \qquad (2M)^2 = 8, \qquad p_a(2M) = 1, \qquad (3M)^2 = 18, \qquad p_a(3M) = 4,$$

there are only three reduced component partitions to consider.

Partition $3M$. If C is irreducible of class $3M$, then its arithmetic genus is 4, so the number of nodes is at most 4.

Partition $2M + M$. Write $C = C_2 + C_1$ with $C_2 \in |2M|$ and $C_1 \in |M|$. The component C_2 has arithmetic genus 1, hence contributes at most one intrinsic node. The component C_1 is smooth rational. Their intersection number is

$$(2M) \cdot M = 4.$$

Each intersection point of C_1 and C_2 is at worst an ordinary node of $C_1 \cup C_2$ (a tangency or higher contact would produce a singularity worse than a node, reducing the node count). Therefore the total number of nodes is at most

$$1 + 4 = 5.$$

Partition $M+M+M$. Each M-component is smooth rational and any two such components meet in

$$M^2 = 2$$

points. Hence the total number of nodes is at most

$$3 \cdot 2 = 6.$$

Thus every reduced divisor in $|3M|$ has at most six nodes. □

Theorem 6.12 (The pure $8A_1$ basket is impossible)**.** *There is no rank-one Gorenstein del Pezzo surface in characteristic* 3 *with basket* $8A_1$*.*

Proof. Assume such an X exists. By proposition 6.1, it has degree 1. Let $f\colon Y \to X$ be its crepant resolution. By proposition 6.4, Y is a degree-one weak del Pezzo surface. By proposition 6.6, the anticanonical ring of X agrees with that of Y.

By proposition 6.7, the anticanonical model $X \cong \mathrm{Proj} R(Y, -K_Y)$ admits a finite separable double cover $\pi\colon X \to Q \cong \mathbb{P}(1,1,2)$, and after resolving Q to $\mathbb{F}_2$, the branch divisor has the form $B = S + C$ with $C \in |3M|$ and $C \cdot S = 0$.

By remark 6.9, the eight A_1-singularities of X correspond, via lemma 6.8, to eight ordinary nodes of C. Therefore C must have at least eight nodes. But by lemma 6.11, a reduced divisor $C \in |3M|$ on $\mathbb{F}_2$ has at most six ordinary nodes. This contradiction proves that $8A_1$ cannot occur. □

7 Proof of the main theorem

Proof of theorem 1.1. Assume for contradiction that X is a rank-one klt del Pezzo surface in characteristic 3 with $N(X) = 8$. By proposition 3.1, its basket is one of the five candidate baskets listed there. The three exotic baskets are impossible by theorem 4.3. The mixed basket is impossible by theorem 5.7. The pure Gorenstein basket $8A_1$ is impossible by theorem 6.12. Thus no rank-one klt del Pezzo surface in characteristic 3 has eight singular points. Therefore $N(X) \leq 7$. □

The public repository accompanying this paper is organized in two layers. The first layer—the scripts and result files cited in appendix C—contains the exact computer-assisted inputs used in the final proof. The second layer records older exploratory branches that are not logically required by the theorem but are retained for provenance and for future machine-assisted work. In particular, the repository preserves the earlier ambient/deformation lanes, quasi-elliptic probes, wild additive-action scans, and the matroid-style structural screens that were useful during search even though they do not appear as formal hypotheses in the final argument.

A The basket-reduction computation

This appendix records the exact data behind proposition 3.1. The point is not to reproduce every line of the scripts in prose, but to expose enough structure and exact counts that a referee can audit the reduction without reverse-engineering the repository.

A.1 Local cyclic types retained by the $N = 8$ screen

The local screen retains the 17 cyclic quotient types listed in table 2. For each type we record the Hirzebruch–Jung chain, the exceptional rank, the local discrepancy correction $\delta_P := K_X^2 - K_Y^2$ contributed by that singularity, and the determinant of the exceptional lattice.

Table 2: Local cyclic quotient types used in the exact basket scan.

r	q	chain	exc-rk	δ_P	det
2	1	[2]	1	0	2
3	1	[3]	1	1/3	3
4	1	[4]	1	1	4
5	1	[5]	1	9/5	5
6	1	[6]	1	8/3	6
7	1	[7]	1	25/7	7
8	1	[8]	1	9/2	8
9	1	[9]	1	49/9	9
10	1	[10]	1	32/5	10
11	1	[11]	1	81/11	11
3	2	[2,2]	2	0	3
5	2	[3,2]	2	2/5	5
8	3	[3,3]	2	1	8
7	2	[4,2]	2	8/7	7
11	3	[4,3]	2	20/11	11
9	2	[5,2]	2	2	9
11	2	[6,2]	2	32/11	11

A.2 From local data to a finite basket list

The scripts implement the following exact finite procedure.

(a) Compute the local invariants in table 2: Hirzebruch–Jung chain, exceptional rank, determinant, local discrepancy correction, and Cartier index.

(b) Form all unordered 8-point multisets from those local types.

(c) For each multiset $\mathcal{B}$, compute the candidate degree

$$K_X^2(\mathcal{B}) = 9 - \sum r_i + \sum \delta_i$$

as in lemma 3.2, and discard every basket with nonpositive degree.

(d) Impose the coarse square-discriminant condition

$$|A_R| m^2 K_X^2 \in (\mathbf{Z}_{>0})^2.$$

(e) On the low-determinant window, run the exact cyclic-complement search in the discriminant group.

(f) Replace the provisional quadratic-form filter by the corrected odd-lattice screen on the discriminant bilinear form and anticanonical complement.

At no stage is there any numerical approximation; all quantities are integral or rational and every search space is finite.

A.3 Exact counts

The exact enumeration and screening pipeline is:

(1) Enumerate all 8-point multisets built from the 17 local types in table 2. This gives

$$735{,}471$$

multisets.

(2) Retain those with positive candidate K_X^2. This leaves

$$732{,}878$$

multisets.

(3) Impose the coarse square-discriminant budget. This leaves

$$4{,}894$$

multisets.

(4) Resolve the small-discriminant window exactly. The corrected exact search decides 45 baskets immediately in the initial determinant window; 24 pass and 21 fail, while two additional historically unresolved baskets are later resolved negatively in the corrected pipeline.

(5) Replace the provisional quadratic-form screen by the corrected odd-lattice screen on the discriminant bilinear form and anticanonical complement. This leaves exactly the five baskets of proposition 3.1.

Two historical corrections are worth recording because the present appendix deliberately uses only the corrected pipeline. First, an earlier quadratic-form implementation falsely rejected the pure $8A_1$ basket because of a bilinear-pairing normalization error (the old v75 rejection). Second, the v78 provisional odd-lattice screen had to be replaced by the corrected v79 bilinear/anticanonical screen. The present paper uses only the corrected output after those repairs.

B Computational ingredients used later in the paper

For convenience we record the exact finite counts used in the later sections.

Exotic baskets

For the normalized exotic blocks of proposition 4.1, the corrected orthogonal real-root counts are:

$$56 \text{ roots for } \tfrac{1}{7}(1,1), \qquad 48 \text{ roots for } \tfrac{1}{7}(1,2), \qquad 32 \text{ roots for } \tfrac{1}{11}(1,2).$$

The repository contains the explicit orthogonality graphs and exact clique computations used in lemma 4.2. In the two chain cases the maximal size of a pairwise orthogonal subset is 4.

Mixed basket

For the mixed basket:

- the polarization solve leaves exactly two orbits L_1 and L_2;
- for L_1, there are 90 roots with $L_1 \cdot r = 0$, 210 roots with $L_1 \cdot r = -2$, and 30,240 transformed noncanonical five-cliques, none compatible with an $A_2 + A_2 + A_1$ subsystem;
- for L_2, there are 128 roots with $L_2 \cdot r = 0$, 177 roots with $L_2 \cdot r = -2$, 13,440 canonical $A_2 + A_2 + A_1$ subsystems split into two Weyl orbits of sizes 3360 and 10080, and 5600 transformed noncanonical five-cliques, all compatible only with the 3360-orbit.

After grouped reduction, there are exactly 16 grouped five-cliques and exactly 7 symmetry orbits.

C Repository map

The repository is intended to make the computer-assisted parts of the proof auditable and reproducible, for both human referees and machine-assisted verification. It is therefore organized in two layers.

(i) The *final proof layer* contains the scripts, JSON summaries, and notes that are actually cited in the paper.

(ii) The *exploratory layer* preserves earlier branches that were useful for search, correction, and provenance, but are not load-bearing inputs to the theorem unless explicitly cited.

Final proof layer

The final theorem depends on the following artifacts.

(A) The basket reduction of proposition 3.1 is recorded in the scripts and outputs corresponding to the v74–v79 chain.

(B) The exotic-basket normalization and root-count computations are recorded in the v94 scripts and output summaries.

(C) The mixed-basket polarization reduction and the global L_2 normalization theorem are recorded in the v95, v96, v97, and v98 artifacts.

(D) The pure $8A_1$ branch-node computation is recorded in the v72 and v99 artifacts.

Exploratory layer

The public repository also preserves three older search lanes because they are useful for understanding the evolution of the argument and for future machine-assisted continuation.

(E1) **Ambient/deformation and Picard-merge lane.** This directory contains earlier projective and higher-dimensional search infrastructure, including matroid-style structural screens, compiled finite-field backends, and the associated handoff notes. These files are not used as theorem inputs in the present paper.

(E2) **Quasi-elliptic/global-probe lane.** This directory records later ambient and quasi-elliptic scans that helped close or deprioritize several branches before the final blowup-lattice proof stabilized.

(E3) **Wild α_3/Artin–Schreier lane.** This directory preserves the additive-action and Artin–Schreier no-go ledgers and associated search scripts. They are historically important because they explain why several characteristic-3 loopholes were investigated and then abandoned.

A future reader should therefore use the following rule: if a file lives in the exploratory layer and is not explicitly cited in the body of the paper, then it is preserved for provenance only and not as a theorem dependency.

The GitHub-ready archive accompanying this draft preserves both the paper files and the machine-readable computation summaries. The public repository URL for this archive is `https://github.com/HiroSakuraba/DelPezzo7`, which points to the reproducibility repository corresponding to this draft.

References

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